

CREDIT CARD FRAUD DETECTION USING AIML

PROJECT ANALYSIS REPORT

Title

Credit Card Fraud Detection using Machine Learning (Random Forest Classifier)

Abstract

Credit card fraud has become one of the most significant financial crimes due to the rapid growth of online transactions and digital payment systems. Financial institutions require intelligent systems capable of identifying fraudulent transactions accurately while minimizing false alarms. This project presents a machine learning-based fraud detection system using the Random Forest Classifier. The model is trained on a highly imbalanced credit card transaction dataset containing both genuine and fraudulent transactions. Various visualization techniques, feature importance analysis, ROC Curve, Precision-Recall Curve, Confusion Matrix, Correlation Heatmap, and SHAP (SHapley Additive Explanations) are employed to evaluate and interpret the model's performance. Experimental results demonstrate excellent predictive performance with an ROC-AUC score of 1.00 and an Average Precision score of 0.95, indicating the effectiveness of the proposed approach.

1. Introduction

The increasing popularity of online banking, digital wallets, and electronic payments has significantly increased the risk of financial fraud. Credit card fraud results in substantial monetary losses for banks, businesses, and customers. Since fraudulent transactions constitute only

a tiny fraction of total transactions, detecting them is a challenging classification problem.

Machine Learning provides automated techniques capable of learning hidden patterns from historical transaction data and identifying suspicious transactions in real time. In this project, a Random Forest classifier is developed to classify credit card transactions as either genuine or fraudulent.

2. Dataset Description

The dataset used is the well-known **Credit Card Fraud Detection Dataset**, which contains anonymized transaction records.

Dataset Information

Attribute	Description
Total Transactions	284,808
Total Features	31
Target Variable	Class
Genuine Transactions	Class = 0
Fraud Transactions	Class = 1

Features

The dataset contains the following variables:

- Time
- V1 to V28 (Principal Component Analysis transformed features)
- Amount
- Class (Target Variable)

Target Classes

Class	Meaning
0	Genuine Transaction
1	Fraudulent Transaction

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis helps understand the characteristics of the dataset before model training.

3.1 Fraud vs Genuine Transactions



Observation

The dataset is highly imbalanced.

- Genuine Transactions $\approx 284,315$

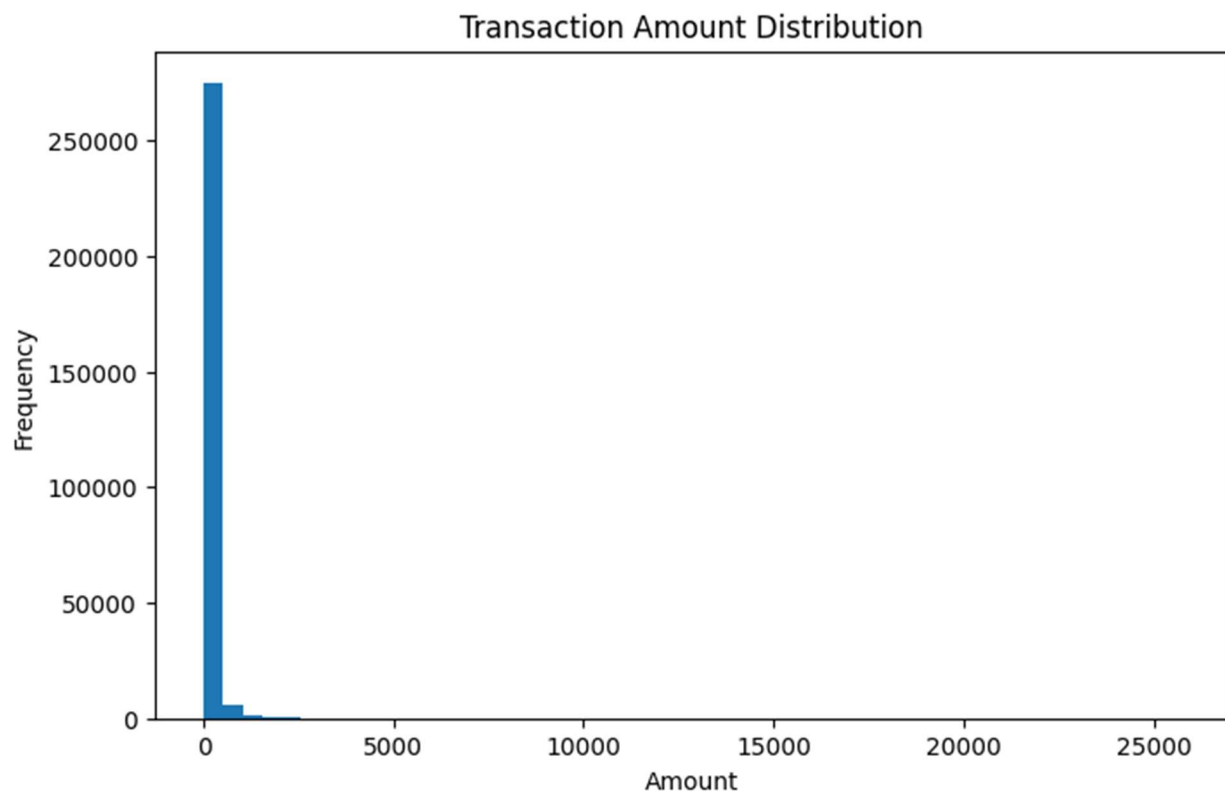
- Fraud Transactions ≈ 493

Fraud accounts for only approximately **0.17%** of all transactions.

Interpretation

This imbalance makes fraud detection a difficult problem because a classifier can achieve very high accuracy simply by predicting every transaction as genuine. Therefore, accuracy alone is not an appropriate evaluation metric.

3.2 Transaction Amount Distribution



Observation

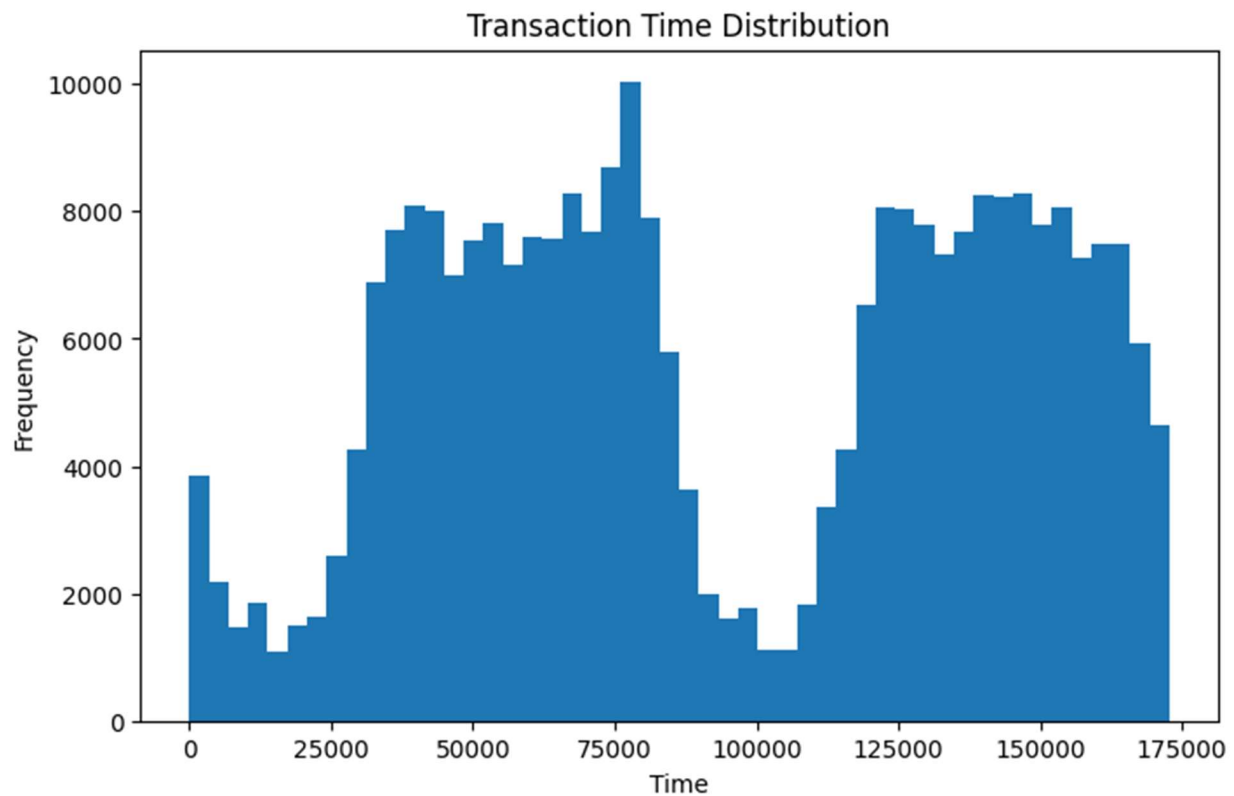
The histogram shows a heavily right-skewed distribution.

Most transactions have relatively low monetary values, while only a few transactions involve very large amounts.

Interpretation

Customers generally perform small-value transactions. Fraudulent activities may occur across both low- and high-value transactions, making transaction amount an important but insufficient feature for fraud detection.

3.3 Transaction Time Distribution



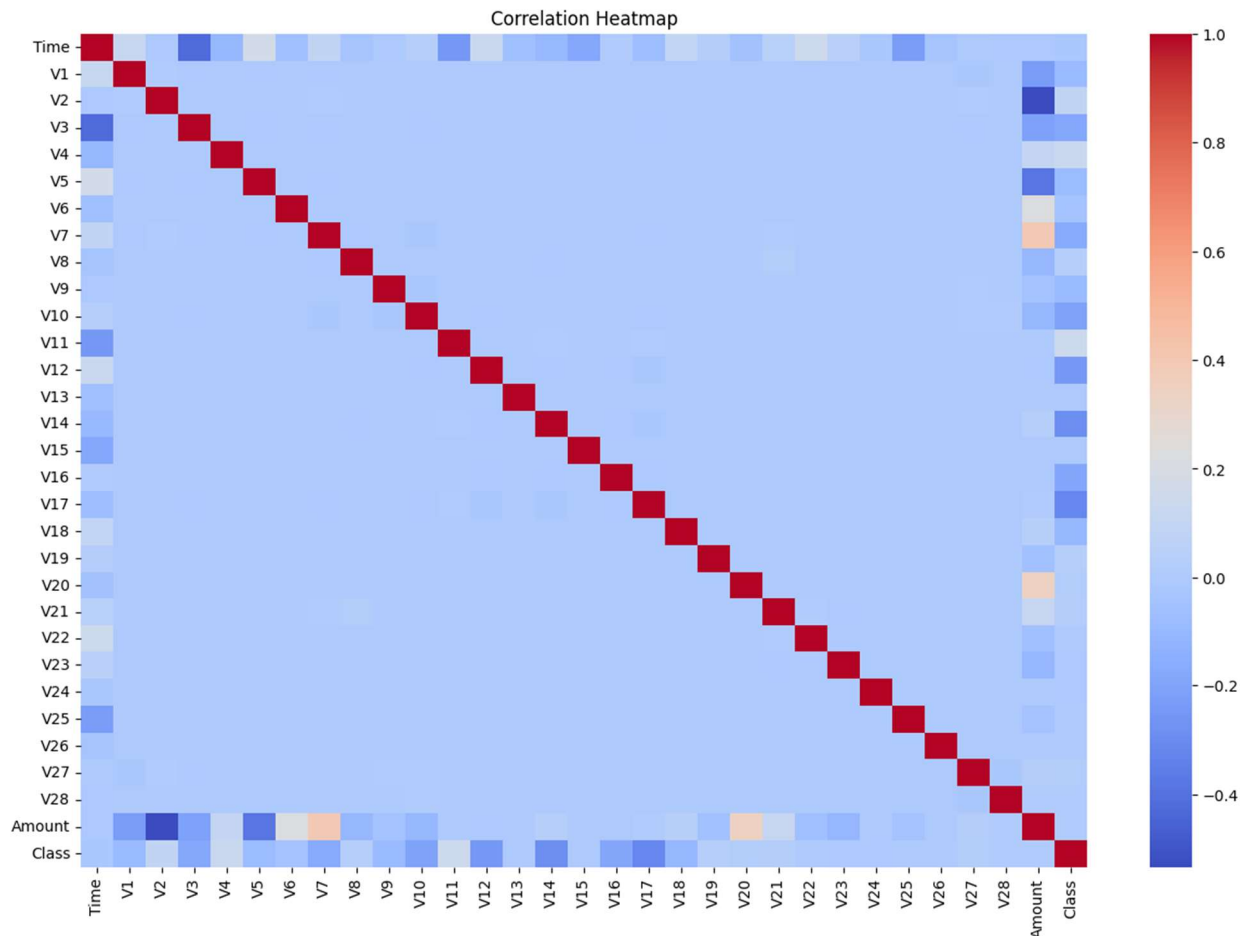
Observation

Transactions are distributed over approximately 172,800 seconds (around two days). The distribution shows varying transaction frequencies during different time intervals.

Interpretation

Transaction time provides useful temporal information that may help identify abnormal purchasing patterns associated with fraudulent behavior.

3.4 Correlation Heatmap



Observation

The correlation matrix reveals that most features have correlations close to zero with each other.

The diagonal values equal one because every variable is perfectly correlated with itself.

Interpretation

Since the V1–V28 variables are generated using Principal Component Analysis (PCA), they are nearly independent. Low multicollinearity improves the performance of many machine learning algorithms, including Random Forest.

4. Machine Learning Model

Algorithm Used

Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting.

Advantages

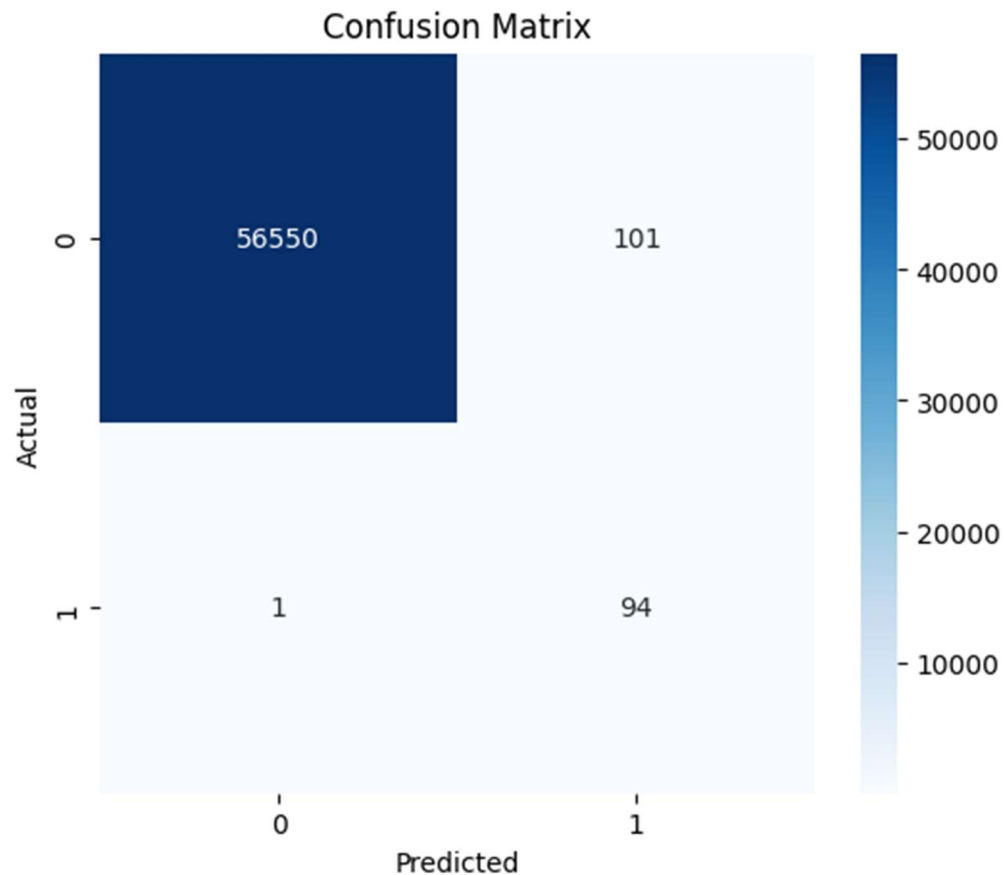
- Handles high-dimensional data
 - Robust to noise
 - Reduces overfitting
 - Provides feature importance
 - Performs well on imbalanced datasets with appropriate evaluation metrics
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5. Model Evaluation

5.1 Confusion Matrix

Actual Class	Predicted Genuine	Predicted Fraud
Genuine	56,550	101

Fraud	1	94
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Interpretation

The model correctly classified:

- 56,550 genuine transactions
- 94 fraudulent transactions

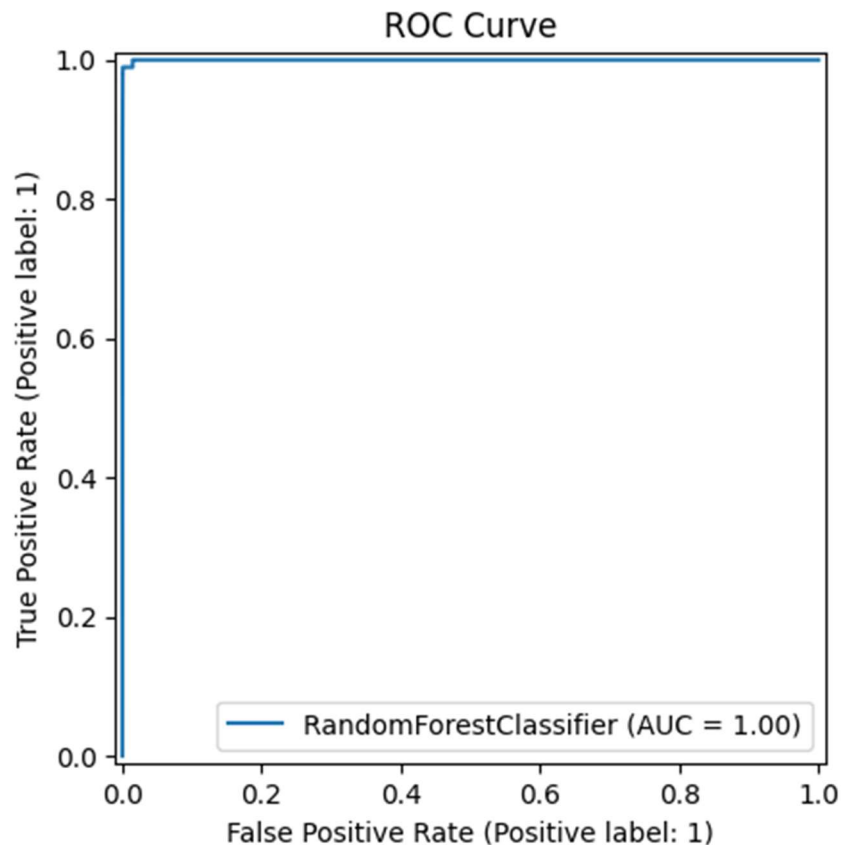
Errors:

- 101 genuine transactions incorrectly classified as fraud (False Positives)
- Only 1 fraudulent transaction missed (False Negative)

Conclusion

The extremely low number of false negatives indicates that the model is highly effective at detecting fraudulent activities while maintaining a low false alarm rate.

5.2 ROC Curve



Result

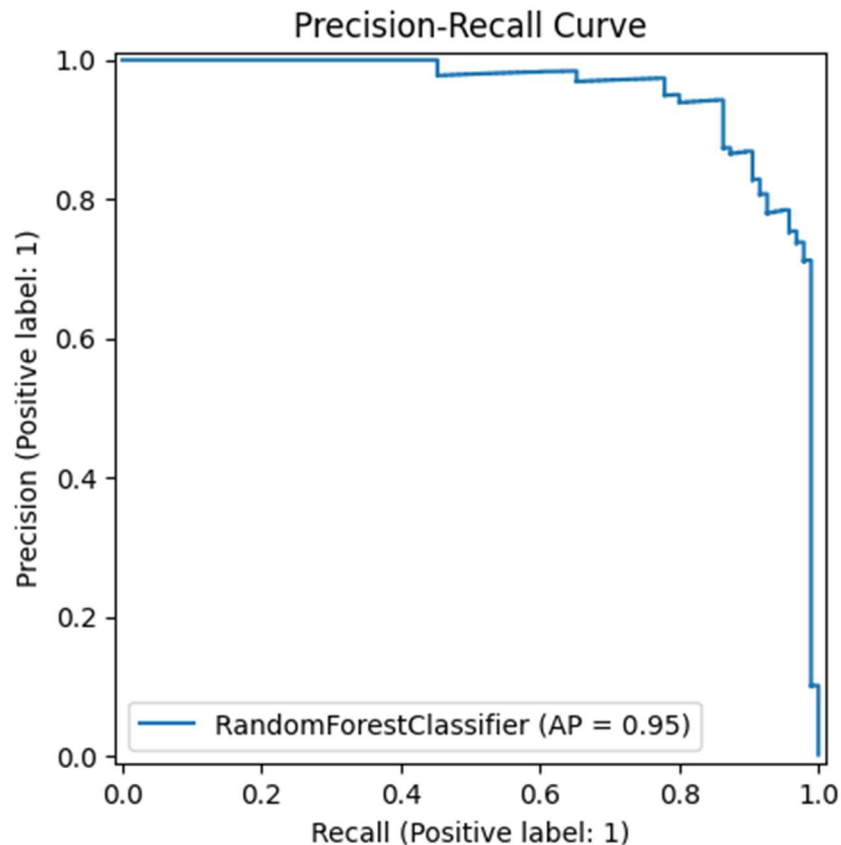
ROC-AUC Score = **1.00**

Interpretation

The Receiver Operating Characteristic (ROC) curve demonstrates the model's ability to distinguish between fraudulent and genuine transactions across different classification thresholds.

An AUC value of 1.00 indicates near-perfect classification performance.

5.3 Precision-Recall Curve



Result

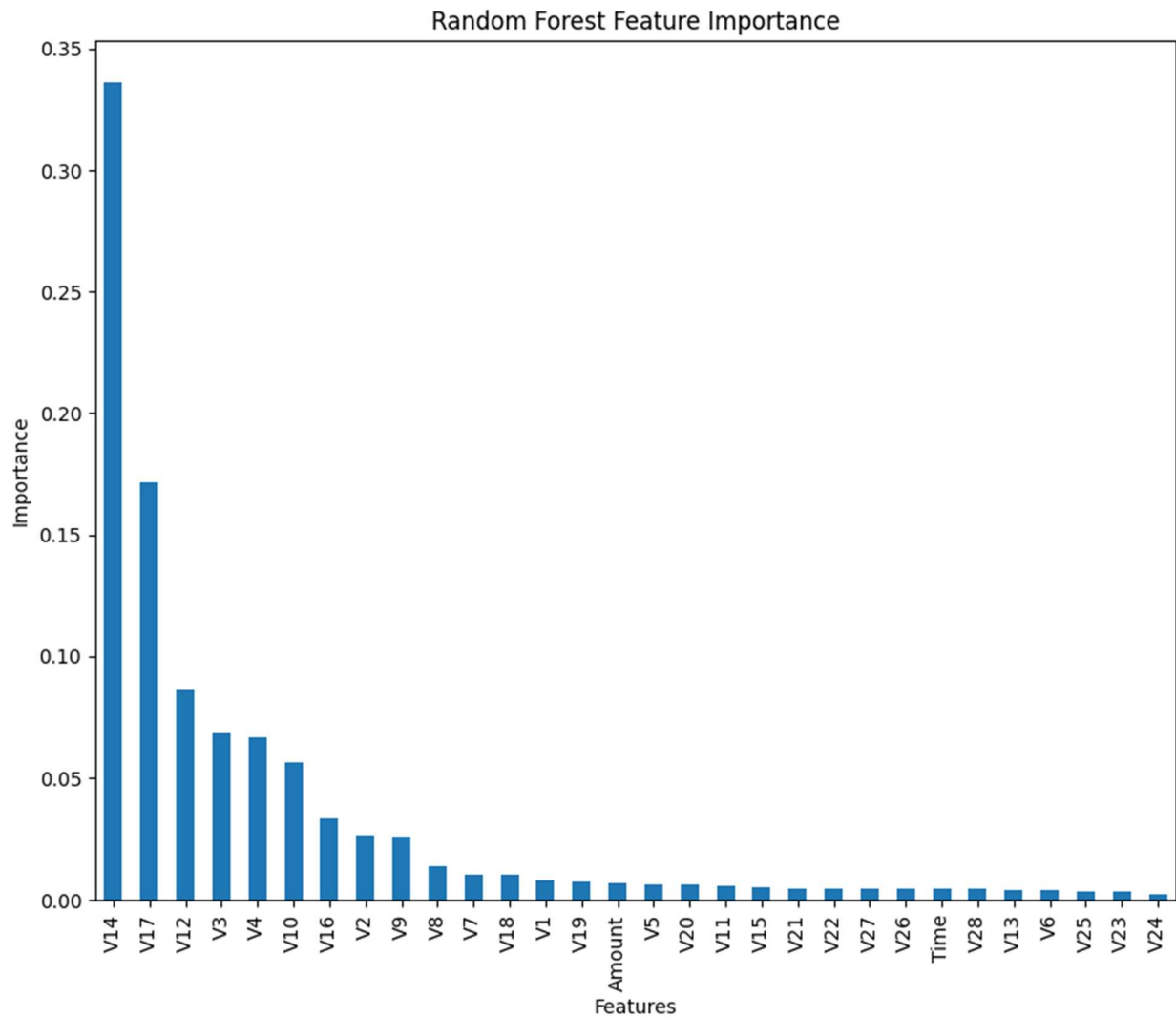
Average Precision (AP) = **0.95**

Interpretation

Because the dataset is highly imbalanced, the Precision-Recall Curve provides a more reliable performance measure than accuracy.

The model maintains high precision while achieving high recall, meaning it detects most fraud cases with relatively few false alarms.

6. Feature Importance Analysis



Random Forest calculates the contribution of each feature toward prediction.

Top Important Features

Rank	Feature
1	V14
2	V17
3	V12

4	V10
5	V4
6	V3
7	V16
8	V2
9	V9
10	V8

Interpretation

Among all variables, **V14** contributes the most toward fraud prediction, followed by **V17**, **V12**, and **V10**. These features contain the strongest fraud-related patterns learned by the Random Forest model.

7. SHAP Explainability

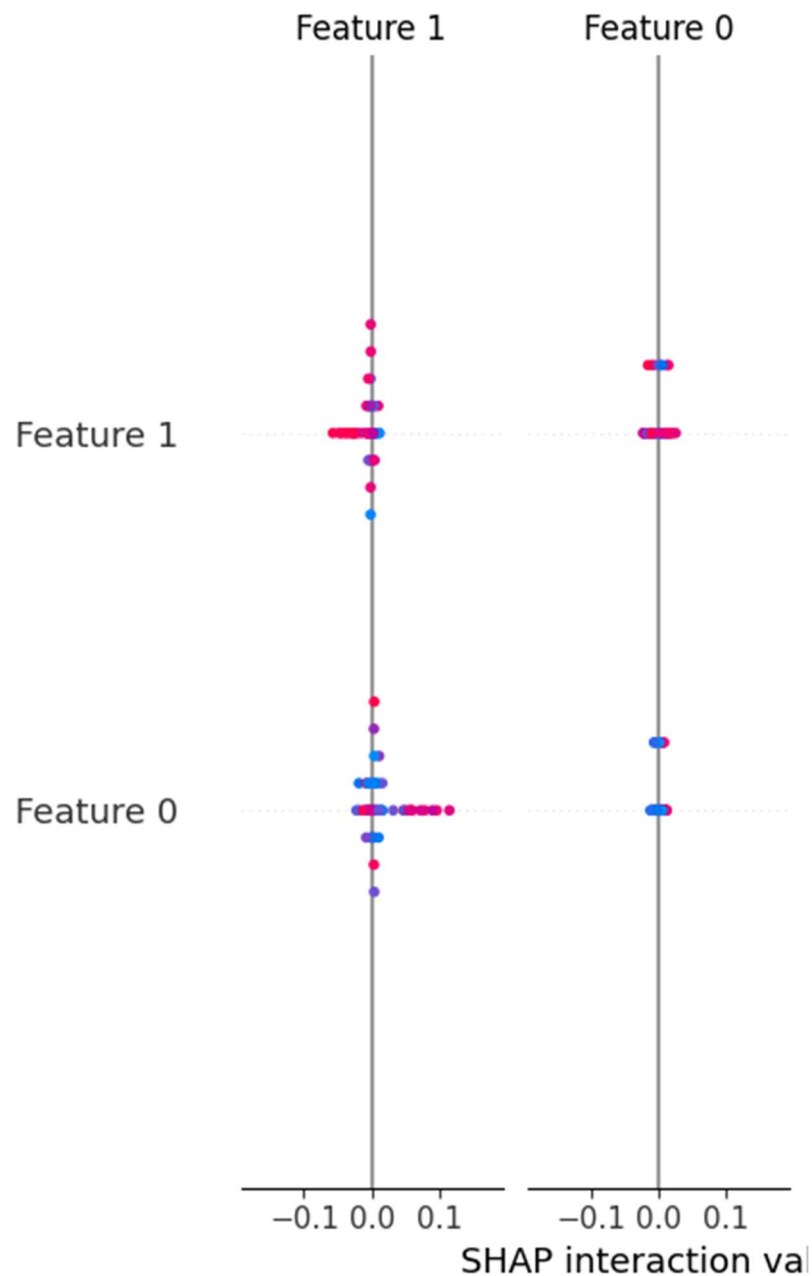
SHAP (SHapley Additive Explanations) provides insight into how each feature contributes to model predictions.

Observation

The SHAP interaction plot indicates that the interaction values between the displayed features remain close to zero.

Interpretation

The model primarily relies on the individual influence of important variables rather than strong interactions between features. Features independently contribute to the prediction process, improving the interpretability of the Random Forest model.



8. Performance Summary

Metric	Value
Dataset Size	284,808
Total Features	31

Genuine Transactions	284,315
Fraud Transactions	493
True Positives	94
True Negatives	56,550
False Positives	101
False Negatives	1
ROC-AUC	1.00
Average Precision	0.95

9. Key Findings

- The dataset is highly imbalanced, with fraudulent transactions accounting for only about 0.17% of all records.
- The Random Forest classifier successfully detects nearly all fraudulent transactions while maintaining a very low false negative rate.
- The ROC-AUC score of 1.00 indicates excellent class separability.
- The Precision-Recall Curve confirms strong performance on the minority class, making the model suitable for fraud detection.
- Feature importance analysis identifies **V14**, **V17**, **V12**, and **V10** as the most influential variables.
- SHAP analysis improves model transparency by explaining individual feature contributions.

10. Advantages of the Proposed Model

- High fraud detection capability
 - Very low false negative rate
 - Robust against overfitting
 - Handles high-dimensional datasets effectively
 - Provides interpretable feature importance
 - Suitable for real-world financial fraud detection applications
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11. Limitations

- The dataset is highly imbalanced, requiring careful evaluation using metrics beyond accuracy.
 - The PCA-transformed features (V1–V28) are anonymized, limiting business interpretation.
 - The model may require periodic retraining as fraud patterns evolve over time.
 - Real-world deployment requires handling streaming transaction data and adapting to concept drift.
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12. Future Enhancements

- Apply advanced ensemble techniques such as XGBoost, LightGBM, or CatBoost.
- Use anomaly detection methods like Isolation Forest or Autoencoders.
- Incorporate real-time streaming analytics for live fraud detection.

- Perform hyperparameter optimization to further improve model performance.
 - Explore explainable AI techniques such as LIME alongside SHAP.
 - Address class imbalance using advanced resampling methods such as SMOTE or ADASYN.
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13. Conclusion

This project demonstrates the successful application of the Random Forest Classifier for credit card fraud detection using a real-world transaction dataset. Despite the extreme class imbalance, the model achieved exceptional predictive performance, correctly identifying **94 out of 95 fraudulent transactions** while misclassifying only **one fraud case**. The **ROC-AUC score of 1.00** and **Average Precision score of 0.95** confirm the model's excellent ability to distinguish between fraudulent and genuine transactions.

Feature importance analysis highlighted **V14, V17, V12, and V10** as the most influential variables, while SHAP analysis provided transparency by explaining how individual features contributed to predictions. Overall, the proposed Random Forest model is accurate, interpretable, and suitable for practical credit card fraud detection systems, offering financial institutions a reliable solution for minimizing fraud-related losses while maintaining efficient transaction monitoring.